# Lecture Scribe Structure

**Lecture_Content.pdf**
PDF

**Structure_Example_Scribe.pdf**
PDF

I'm preparing lecture notes for exam revision and want them to follow a consistent structure throughout the semester for all my lecture scribe.

For that I am sharing you two pdf files.
--> The first PDF file consists of the lecture PDF for which you need to make a complete lecture scribe.
--> The second PDF file consists of the example of structure (template) of the lecture scribe, which is provided only to demonstrate the structure of how the scribe must look like it's formatting style and all.
Do not copy or paraphrase any content from the example scribe.
NOTE: The content of this example file must not be included in the final scribe (only refer to the structure). It is only given to know that how to scribe must be organized.

The goal of this scribe is to help with exam revision.
The scribe must cover these points properly:
- Cover the entire lecture PDF.
- It should be written as exam revision material style.
- Include all the definitions, formulas, equations, derivations, proofs, and all the examples only if they are in the lecture PDF, don't introduce new content.
- Please stick only to the content which appears in the lecture PDF.
- And give derivations for only those which are given, don't include extra or new things.
- And also don't omit any concept from the PDF.
- Maintain a proper academic note taking style similar to well-written human lecture notes.
- Also maintain the order of the topics as presented in the lecture PDF.

How the output should be:
- Follow the structure example file for template of the scribe and do not include any content of it.
- Do not include any content from the example scribe.
- Also please do not include anything in the output which includes any reasoning or explanation of how the scribe was created.

Start generating the lecture scribe only after carefully analyzing both the files.

# CSE400 – Fundamentals of Probability in Computing

## Lecture 5: Bayes' Theorem, Random Variables, and Probability Mass Function (PMF) (January 20, 2026)

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## 1. Overview of the Lecture

This lecture covers three connected ideas used frequently in probability for computing:

- **Bayes' Theorem** (as weighted conditional probabilities + formal law of total probability + Bayes formula)
- **Random Variables** (why we define them and how their distributions are described)
- **Probability Mass Function (PMF)** for discrete random variables (definition + examples)

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## 2. Key Definitions and Terminology

### 2.1 Mutually Exclusive (Disjoint) Events

If events cannot happen together, they are **mutually exclusive**. In the lecture, $AB$ and $AB^c$ are disjoint.

### 2.2 Random Variable

A **random variable** is a real-valued function defined on the sample space; its value is determined by the outcome of the experiment.

### 2.3 Distribution of a Random Variable

For each possible value $a$, the quantity $\Pr[X = a]$ gives the probability that the random variable equals $a$. The lecture notes that the distribution can be visualized using a **bar diagram** where bar height at $a$ equals $\Pr[X = a]$.

### 2.4 Discrete Random Variable

A random variable is **discrete** if it can take **at most a countable** number of possible values.

### 2.5 Probability Mass Function (PMF)

For a discrete random variable $X$ with range $R_X = \{x_1, x_2, \dots\}$, the function assigning probabilities to each possible value is called the **PMF** of $X$.

### 2.6 Apriori vs Posteriori Probability (as stated in lecture)

- $\Pr(B_i)$: **apriori** probability (from presupposed/self-evident model)
- $\Pr(B_i \mid A)$: **posteriori** probability (after observing event $A$)

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## 3. Explanation of Core Concepts and its Understanding

### 3.1 Bayes' Theorem as a Weighted Average of Conditional Probabilities

Key idea: write $A$ as

$$A = AB \cup AB^c$$

so outcomes in $A$ are either "in $A$ and $B$" or "in $A$ and not $B$".

Because $AB$ and $AB^c$ are mutually exclusive,

$$\Pr(A) = \Pr(AB) + \Pr(AB^c).$$

Then using conditional probability forms (as shown in the lecture), we get:

$$\Pr(A) = \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)(1 - \Pr(B)).$$

This is interpreted in the lecture as: $\Pr(A)$ **is a weighted average of conditional probabilities**, weighted by how likely the conditioning events are.

### 3.2 Law of Total Probability

The lecture introduces the law of total probability as a formal tool.
If $B_1, \dots, B_n$ are mutually exclusive and partition the sample space, then:

$$\Pr(A) = \sum_{i=1}^{n} \Pr(A \mid B_i) \Pr(B_i).$$

(Referenced in lecture as “law of total probability [Formula 3.4]”.)

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### 3.3 Bayes Formula (Formal Bayes’ Theorem)

Using the relationship (stated in the lecture):

$$\Pr(AB_i) = \Pr(B_i \mid A) \Pr(A),$$

the lecture arrives at **Bayes Formula [Proposition 3.1]**.

Standard form consistent with the lecture’s setup:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i) \Pr(B_i)}{\sum_{j=1}^{n} \Pr(A \mid B_j) \Pr(B_j)}.$$

Also recall: $\Pr(B_i)$ is apriori and $\Pr(B_i \mid A)$ is posteriori as described.

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### 3.4 Random Variables: Why we define them

Instead of tracking the full outcome, we often track a **function** of the outcome (examples given: sum of two dice; number of heads). These functions are random variables.

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### 3.5 Distribution Visualization (Bar Diagram idea)

- x-axis: values the RV can take
- bar height at value $a$: $\Pr[X = a]$
- these probabilities come from the probability of the corresponding event in the sample space

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### 3.6 PMF Concept (Discrete Case)

A discrete random variable has countable possible values; PMF assigns probability to each single value.

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## 4. Important Formulas and Mathematical Expressions

### 4.1 Event Decomposition

$$A = AB \cup AB^c$$

### 4.2 Weighted Average Form (Two-case total probability)

$$\Pr(A) = \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)(1 - \Pr(B))$$

### 4.3 Law of Total Probability (General partition)

$$\Pr(A) = \sum_{i=1}^{n} \Pr(A \mid B_i)\Pr(B_i)$$

### 4.4 Bayes Formula (Proposition 3.1)

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i)\Pr(B_i)}{\sum_{j=1}^{n} \Pr(A \mid B_j)\Pr(B_j)}$$

### 4.5 Conditional Probability used in examples

$$\Pr(A \mid A_1) = \frac{\Pr(AA_1)}{\Pr(A_1)}$$

### 4.6 Random Variable probabilities (distribution statement used repeatedly)

$$\Pr[X = a]$$

(used for bar diagram interpretation)

### 4.7 PMF example family (given)

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots, \ \lambda > 0$$

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## 5. Derivations / Proofs (If Present in Lecture)

### 5.1 Derivation: "Weighted Average of Conditional Probabilities"

**Start:**

$$A = AB \cup AB^c$$

**Step 1 (mutual exclusivity):**

$$\Pr(A) = \Pr(AB) + \Pr(AB^c)$$

**Step 2 (convert to conditional probabilities):**

$$\Pr(AB) = \Pr(A \mid B)\Pr(B), \qquad \Pr(AB^c) = \Pr(A \mid B^c)\Pr(B^c)$$

and $\Pr(B^c) = 1 - \Pr(B)$. This yields the lecture's final form:

$$\Pr(A) = \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)(1 - \Pr(B)).$$

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### 5.2 Derivation: Determining the constant $c$ in $p(i) = c\frac{\lambda^i}{i!}$

**Given PMF form:** $p(i) = c\frac{\lambda^i}{i!}$, $i = 0, 1, 2, \dots$.

**Start with PMF property:**

$$\sum_{i=0}^{\infty} p(i) = 1$$

So,

$$\sum_{i=0}^{\infty} c\frac{\lambda^i}{i!} = 1 \quad \Rightarrow \quad c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1.$$

**Use the series identity (as used in the lecture's worked slide):**

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

Therefore,

$$ce^{\lambda} = 1 \quad \Rightarrow \quad c = e^{-\lambda}.$$

(This matches the final constant shown in the lecture's worked example slide.)

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## 6. Worked Examples / Applications

### 6.1 Example 3.1 (Part 1/2): Insurance (Total Probability idea)

**Problem (given):** Accident-prone class vs not accident-prone class;

- Pr(accident within 1 year | accident prone) = 0.4
- Pr(accident within 1 year | not accident prone) = 0.2
- Pr(accident prone) = 0.3
  Find Pr(accident within 1 year).

**Solution (as in lecture):** Let $A_1$ = accident within a year, $A$ = accident prone.

$$\Pr(A_1) = \Pr(A_1 \mid A)\Pr(A) + \Pr(A_1 \mid A^c)\Pr(A^c)$$

$$= (0.4)(0.3) + (0.2)(0.7) = 0.26.$$

**Final answer:** $\boxed{0.26}$.

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### 6.2 Example 3.1 (Part 2/2): Insurance (Bayes / Posterior probability)

**Problem (given):** Given that the policyholder had an accident within a year, find the probability they are accident prone.

**Solution (as in lecture):**

$$\Pr(A \mid A_1) = \frac{\Pr(AA_1)}{\Pr(A_1)} = \frac{\Pr(A)\Pr(A_1 \mid A)}{\Pr(A_1)} = \frac{(0.3)(0.4)}{0.26} = \frac{6}{13}.$$

**Final answer:** $\boxed{\Pr(A \mid A_1) = 6/13}$.

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### 6.3 Example 3.2: Three cards (Bayes reasoning with "equally likely outcomes")

**Problem (given):** Three cards: RR, BB, RB. Pick one at random, place it down. If the upper side is red, find probability the other side is black.

**Setup (as in lecture):** Let events be RR, BB, RB and let $R$ be "upturned side is red". The lecture sets up:

$$\Pr(RB \mid R) = \frac{\Pr(R \mid RB)\Pr(RB)}{\Pr(R \mid RR)\Pr(RR) + \Pr(R \mid RB)\Pr(RB) + \Pr(R \mid BB)\Pr(BB)}.$$

With:

- $\Pr(R \mid RB) = 1/2$, $\Pr(RB) = 1/3$
- $\Pr(R \mid RR) = 1$, $\Pr(RR) = 1/3$
- $\Pr(R \mid BB) = 0$, $\Pr(BB) = 1/3$

**Final result (lecture conclusion):**
The probability equals $1/3$. The lecture explains the common wrong guess $1/2$ happens by incorrectly assuming the remaining possibilities are equally likely; correct reasoning counts equally likely "sides" outcomes and shows only one favorable outcome (R3) among three red-up outcomes (R1, R2, R3).

**Final answer:** $\boxed{1/3}$.

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### 6.4 Random Variable Example 1: Tossing 3 fair coins

**Problem (given):** Toss 3 fair coins. Let $Y$ be the number of heads.

**Values and probabilities (as listed):**

- $\Pr(Y = 0) = 1/8$
- $\Pr(Y = 1) = 3/8$
- $\Pr(Y = 2) = 3/8$
- $\Pr(Y = 3) = 1/8$

Also, since $Y$ must take one of these values, total probability sums to 1 (stated in the lecture).

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### 6.5 PMF Example Mentioned: Two Independent Tosses of a Fair Coin

The lecture includes a slide titled **"Example: Two Independent Tosses of a Fair Coin"** under PMF.
(Only the example heading is present in the lecture PDF text at that point.)

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### 6.6 PMF Example: $p(i) = c\frac{\lambda^i}{i!}$ — find $\Pr(X = 0)$ and $\Pr(X > 2)$

**Given:** $p(i) = c\frac{\lambda^i}{i!}$, $i = 0, 1, 2, \ldots, \lambda > 0$.

**Step 1: Find** $c$ (from derivation above): $c = e^{-\lambda}$.

**Step 2: Compute** $\Pr(X = 0)$

$$\Pr(X = 0) = p(0) = c\frac{\lambda^0}{0!} = c = e^{-\lambda}.$$

**Step 3: Compute** $\Pr(X > 2)$
Use complement form (as shown conceptually in the worked slide):

$$\Pr(X > 2) = 1 - \Pr(X \leqslant 2) = 1 - (\Pr(X = 0) + \Pr(X = 1) + \Pr(X = 2)).$$

Where:

$$\Pr(X = 1) = p(1) = e^{-\lambda}\frac{\lambda}{1!} = e^{-\lambda}\lambda, \quad \Pr(X = 2) = p(2) = e^{-\lambda}\frac{\lambda^2}{2!} = e^{-\lambda}\frac{\lambda^2}{2}.$$

So,

$$\boxed{\Pr(X > 2) = 1 - e^{-\lambda}\left(1 + \lambda + \frac{\lambda^2}{2}\right)}.$$

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## 7. Important Observations and Results

- **Weighted-average interpretation:** $\Pr(A)$ can be expressed as a weighted average of $\Pr(A \mid B)$ and $\Pr(A \mid B^c)$, with weights $\Pr(B)$ and $1 - \Pr(B)$.
- **Common Bayes pitfall (cards example):** After seeing "red", the remaining cases are not automatically equally likely; you must account for the correct underlying equally-likely outcomes.
- **Random variables shift focus:** we track a numerical quantity derived from outcomes (sum, count, etc.), not necessarily the full outcome.
- **Distribution view:** A distribution can be represented via bars: value vs $\Pr[X = a]$.
- **Discrete RV → PMF:** Discrete means countable outcomes, enabling PMF assignment to each value.

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## 8. Quick Revision Points

- **Decompose event:** $A = AB \cup AB^c$.
- **Two-case total probability:**
  $\Pr(A) = \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)(1 - \Pr(B))$.
- **Law of total probability (partition):**
  $\Pr(A) = \sum_i \Pr(A \mid B_i)\Pr(B_i)$.
- **Bayes formula:**
  $\Pr(B_i \mid A) = \dfrac{\Pr(A \mid B_i)\Pr(B_i)}{\sum_j \Pr(A \mid B_j)\Pr(B_j)}$.
- **Insurance example results:**
  $\Pr(A_1) = 0.26$, and $\Pr(A \mid A_1) = 6/13$.
- **Cards example result:** $\Pr(\mathrm{RB} \mid R) = 1/3$.
- **3-coin RV distribution:** $1/8, 3/8, 3/8, 1/8$ for $Y = 0, 1, 2, 3$.
- **PMF family:** If $p(i) = c\frac{\lambda^i}{i!}$, then $c = e^{-\lambda}$, $\Pr(X = 0) = e^{-\lambda}$, and
  $\Pr(X > 2) = 1 - e^{-\lambda}\left(1 + \lambda + \frac{\lambda^2}{2}\right)$.

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### Class Participation Note (from lecture)

The lecture ends with a class participation prompt to switch to Campuswire.

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**Template reference used only for structure (not content):**